

Research Report

Network Encryption and Traffic Analysis

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July 16, 2026

Abstract

Modern encryption protocols are practically unbreakable, this ensures that the content of the communication is protected, but metadata and traffic patterns can still be exploited to infer information. The research examines how protocols such as TLS, IPsec, HTTPS secure network communications. We then investigated different techniques used in network analysis to extract intelligence without breaking encryption such as Leakage Vectors, Inversion attacks, Membership inference attacks. Many of these exploits utilize machine learning to learn the patterns of packets and user data, the process of learning patterns can result in vectorization of personal data, its necessary to ensure that no leaks are the result of improper training of machine learning models. Military or government communications require additional network security, the report answers why encryption alone is not enough in this context.

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1 Introduction

1.1 The Web, Privacy and Confidentiality

In the 1970s ARPANET was born, a packet based network that changed how computers were seen, from arithmetic devices to communication devices. This change in perspective is historical as it changed how computers were used and how we interacted with them, and most importantly, physical isolation stopped being a form of security.[1] Until the 1990s networks were still isolated and adoption was low, but with the birth of protocols such as HTTP global adoption followed. Today, computers and telematic communications are everywhere and, most importantly, they are used for sensitive information. Both our personal data such as, behavior, personal sphere, personal identity but also our sensitive data such as work communications, bank transactions, finance information travel through the internet. Considering the scale and heterogeneity of the world wide web it's not feasible to ensure the security of the channels, therefore it's imperative to ensure a reasonable level of security through encryption ensuring privacy and confidentiality.

1.2 The limitations of encryption

Modern public encryption thanks to protocols such as HTTPS or encryption algorithms such as RSA is practically unbreakable as far as the public domain is concerned, therefore the actual data that has been sent through a channel cannot be seen or decrypted by a malicious actor. This does not mean that encryption is always enough, even if explicit data is protected a lot can still be inferred. The techniques used as discussed in the report vary but all exploit the fact that encrypted data still has to travel as packets: the size, the timestamp, the frequency and the structure of the packet can all be used to infer information about the original data even without breaking encryption. This limitation of encryption is not only relevant to civilians communications, military communications are very sensitive information that states guard as closely as possible to ensure military superiority [2], therefore both defending and attacking military networks is key, managing to infer a rough enemy position might be critical information to assert dominance over the battlefield.

1.3 The double edge of machine learning

A powerful tool used to analyze network traffic is machine learning; it can be used to learn patterns invisible to the human eye allowing for precise inference of the original data up to 98% [3]. This process of learning is achieved through the utilization of vectors and embeddings, allowing the model to categorize information. With well structured datasets the model can retain precision even when the environment changes. But the ability of machine learning models to vectorize information is a double-edged sword; as they become a potential source of information leakage. A sensitive attribute attack or a membership inference attack can reveal sensitive information used during the training of the model.

1.4 Report objective

As we said Network traffic protection is very high priority in the modern interconnected world, encryption has been used for a long time as the main tool to protect our sensitive information but, as we discussed and will analyze more in depth later, its often not enough to ensure complete protection. The objective of this report is therefore to give a rough overview of the

tools and protocols used to ensure network encryption and how metadata can be exploited by malicious attackers to gather information. Lastly we will explore how this topic concerns military networks and if encryption is enough to protect them.

2 Background

2.1 The evolution of encryption

Encryption is the process of turning the original plain message into cipher text, an unrecognizable message different from the starting one. The reverse of encryption is decryption, and the entire process is called cryptography. In general cryptography focuses on privacy and the security of information based on data integrity, confidentiality and authentication. [4]

To secure the exchange of information over the Web, the HTTPS protocol has been created to provide a cryptographic protection. HTTPS is based on other protection protocols that have been improved; it basically runs the HTTP protocol, which is at the center of the Web but does not provide confidentiality or integrity, on top of cryptography protocols, such as SSL and TLS, that allow bidirectional encryption in communication.

Because of the very different levels of security, it is believed that the HTTPS will replace mostly HTTP protocols, also thanks to initiatives and actions taken by the major browser vendors to promote the importance of security. [5]

2.2 Protocols

As it is briefly explained in the paragraph above there are different protocols that contribute to security on the Web. The two more widely used are TLS (Transport Layer Security) and IPsec (Internet Protocol Security).

TLS aims to ensure end-to-end security protection against active attacks, it is a protocol that runs between a client and a server. In the beginning it was only used in web browsers for HTTPS connections, now it is used on a wider scale: e-commerce transactions, VPNs, and mobile apps for Android and iOS. It consists of the Handshake Protocol, which focus is on key establishment and authentication, and the Record Protocol, which allows a secure delivery of the data. TLS has been the subject of many research analysis in order to establish its level of security. If on the one hand, the Record Protocol has been demonstrated to be a secure encryption in many studies, the Handshake Protocol still has some flaws. The main weakness is the interaction between the Handshake and the Record protocols: the application key is used to encrypt the last two messages of the Handshake Protocol. [6]

IPsec is a network protocol that permits the encryption and authentication of data at the IP layer, regardless of the kind of data or function that is being transmitted. In a 1999 evaluation [7] quite a few criticisms were brought up on the process and the results of IPsec. The disappointment on how it was developed was a shared opinion among the people involved in it. But even with the critics it was the best IP security protocol available, in comparison to all the protocols in the market, IPsec seemed to be one with the better performance. Their main criticism was its complexity; it had too many options and too much flexibility, this results in an important weakness when it comes to security. Another criticism was raised on encryption and authentication, it needs to be applied to all the data that can be used to determine the meaning of a packet, authentication needs to be applied to the ciphertext but also to the decryption key.

2.3 The metadata

(What encryption leaves behind: packet sizes, time stamps, directional flow, and IP routing information.) In the paragraphs above we have analyzed ways of encrypting data when exchanged between two or more parties, however some data still needs to be protected: the metadata. A lot of information can be leaked through data records and association, which is why limiting access and data masking becomes as important as encryption. The access to the data needs to be controlled through access requests which are checked and then granted or denied depending on the outcome of the controls. Other techniques used are disassociation techniques which permit the anonymity of the user and its depersonalization to maintain privacy. Sometimes instead of anonymization, which is a non-reversible process, it is used pseudonymization, which is a similar technique but the original identifying information is not deleted but only separated from the data, which is not associated to the pseudonym. Another way to shield sensitive data is encryption again, in this case the key is initially possessed only by the owner. [8] its important to

3 Overview of the problem/evaluation

3.1 Network traffic analysis

The network traffic analysis is an important part for developing control schemes and distinguish a malicious packet from a normal one. Prediction in network traffic is important and there are two categories: long and short period predictions. The long period prediction allows more detailed forecast, it helps to determine future capacity needs and allows better planning and decision taking. Short period predictions can be used for congestion control and a better resource management. [9]

Nowadays network traffic analysis has become very important for monitoring the traffic. Previously, only a few networks devices were kept monitored since the bandwidth was just about 100 Mbps. Now the speed has increased to more than 1 Gbps and more networks have been created, so it is clear how this science has to keep evolving. There are various techniques for network traffic analysis, the typical approach consists of a first processing and inspection of the traffic and then an analysis.

Network Traffic Processing One of the techniques that allow processing a large amount of data is anomaly detection, this technique aims to identify suspicious or unusual activity or behavior. There is a model that represents the 'normal network activity' and then all network flow is analyzed based on the model, it monitors traffic load in order to recognize the triggers to a significant traffic volume change. These systems are built in two phases: the training phase (on a constructed model) and the testing phase (on current traffic). Most of these systems detect anomalies using machine or deep learning techniques.

Another technique is the signature-based network inspection which mostly uses DPI (Deep Packet Inspection) techniques. This techniques is now used in different layer of the OSI (Open Systems Interconnection), unlike earlier which was only used on packet headers. Nowadays protocols are so complex and obfuscated that it is necessary to inspect all layers. Using DPI allows to identify different styles of content and streaming them based on their type; the same suspicious and malicious traffic is identified and then analyzed. However, these DPI based approaches are only able to process unencrypted data because they extract important information from the network packet payload content. These techniques expect unencrypted data which nowadays is not realistic so other methods have been investigated. [10]

Network Encryption Analysis This paragraph discusses approaches for collecting useful side-channel information from network flows despite their full encrypted payloads. Quite some information can be extracted only from the type of traffic or the application it's coming from; some pioneers techniques, like BLINC or Bernaille et al., proposed methods that were able to classify network traffic only based on host behavioral patterns or the sizes of the packets in a SSL connection. These methods had an accuracy of 85%. Some studies and investigations have been made on identifying traffic that mobile application produce, an experiment based their approach on packet sizes. Applications like AppScanner are able to automatically generate applications fingerprints in encrypted traffic using a supervised learning algorithm with an accuracy of 99% on the profiled apps. FLOWPRINT analysis network traffic introducing an approach that combines destination-based clustering, browser isolation and pattern recognition; it has an accuracy of 89-90% even for application updates, newly encountered application and unknown ones. Other researches proposed the use of a recurrent neural network (RNN) with a convolutional neural network (CNN) for traffic classification, these algorithms used with this approach do not require remodelling when applied to new models. [10]

User Identification In this paragraph we will analyze how data can be extracted from a user's actions even when communication is encrypted through the use of machine learning techniques.

Various systems are proposed to analyze encrypted traffic of Android and iOS devices, they mainly use IP packet fields, packet sizes, direction and timing of the exchanges. Through machine learning they analyze numerous applications such as Gmail, facebook, Twitter, Dropbox to identify different actions. These studies show that eavesdroppers are able to gain fine-grained information such as the size of messages or even the language used for the exchange. A system named CUMMA is able to identify the data type (e.g. photo, video, audio...) based on the packet lengths and time delays in WhatsApp and WeChat exchanges with a 96-97%. Another more dangerous study showed that it is possible to identify financial transactions at the trading stage analyzing the encrypted traffic through the data from mobile payment apps, identifying specific actions and the steps within.

There are also works that are able to decode encrypted VoIP (Voice over Internet Protocol) conversations or video streams: audio is encoded using variable bit rates which means the length of the packets can be used to recognize standard words or phrases with a 50% accuracy. On the other hand video is decoded due to specific 'burst patterns' in video streams, an attacker can gain information using flow attributes such as bytes and packet flows per second. [10]

3.2 Information Leakage Vectors

Vectors containing information are created through embeddings; these are mathematical functions that map raw object to real valued vector with the aim of gaining important meanings about the objects. Embeddings are mostly used in Machine learning algorithms to transfer learning: it has resulted to be very effective and useful for many natural language processing for example paraphrasing, response suggestion, retrieving information, text classification and answering questions. Other successful applications are the ones on social networks, movie feedback, locations...

These services rely more on high quality and sensitive data than improved training algorithms and architectures. There is an interesting study [11], motivated by the widespread application of these models, tries to gain a better understanding of how, during the stages of training, information about sensitive data may be captured and then leaked. May they capture infor-

mation about the authors? What are the risks of embeddings? They studied three different type of attacks on embedded models.

Embedding inversion attacks The goal of these attacks is to recover the input text x^* having access to the function Φ and the embedded vectors $\Phi(x^*)$. In the study they analyzed two scenarios, one where it is given access to the parameters and the architecture of the Φ function (white-box) and the other one where attacks are only based on auxiliary data (black-box).

The initial approach when having white box access could be to create every possible combination of words and test which one matches, but there is a computational problem, it is too complex. The solution they proposed is a continuous relaxation of the sequential word input. This approach generates a continuous variable per each word, then based on the highest probability words forms the final text. The attacker can create a small auxiliary dataset to train the model and then retrieve the information with better accuracy.

In the other scenario, the adversary only has access to the embedding model Φ , in this case they do not search for the most likely input but the adversary learns an inversion model that gets in input $\Phi(x)$ and directly output words in the sequence x . The inversion model is trained with an auxiliary data set to maximize the prediction. In order to achieve that it can be used a multi-label classification, with the focus on single words, or a multi-set prediction, which predicts words based on the the set of words already predicted.

Sensitive attributes attacks When the embedding is user related, the vector also captures information about the user and it usually contains more information than just the raw input. In many applications this user related information is used to give the user more personalized services, but it can reveal sensitive information that the user probably did not intend to share. In supervised learning, sensitive attributes can be revealed but they are used as internal features for the learning task; in unsupervised learning the leakage of sensitive information is less controlled. When the model is trained on text unsupervised, it groups vectors that are similar together and stores away different vectors. This way, since a person tends to write with the same style, vocabulary and context the user can be identified.

Membership inference attacks In the paragraph before the focus was on input privacy but there is another important aspect of privacy in ML models. That is the 'training data privacy', what amount of information used during training is leaked by the model. In this study [11] to measure the leakage they use Membership Inference Attack (MIA), the goal is to find out whether a piece of data (words, a phrase or a sentence) has been used to train the model. This permits to discover if the model was trained on private logs or a proprietary code, which would reveal a privacy leak. In the simulations run by the authors they check similarities in the vectors embeddings of the model and the data from a target user, if the similarity scores are too high than the model was trained on that specific data. For example attacks can be made against next word prediction model used to train keyboard apps, but this will be analyzed more in dept in the following paragraph.

3.3 Military Internet of Things

The MIoT has become a new research area in military intelligence and one of the main challenges of such a IoT environment is to acquire, analyze and merge a big amount of data from different devices and spread them in a secure and reliable manner. In this work [12] the authors focus on data security; ensuring secure management of data through the path from the origin to the receiver and implementing precise access control mechanisms. The

problem of data security mostly arises when the military has to interact with other institutions and organizations, share some resources and exchange information, when military systems become interconnected ensuring data security becomes a necessity. The MIoT consists of various layers, the data acquisition and short range wireless communication (such as sensors, GPSs, wearable computer technologies), the communication layer that consist of the transmission of the information received from the previous layer, the data management layer and the intelligent service layer. Every layer needs to follow certain requirements: end to end encryption, authentication and access control, integrity verification, interoperability in federated environments and data-centric security.

4 Examples

In this section we will analyze a couple of examples of applications or attacks based on some methods explained in the previous paragraphs.

4.1 Information Leakage in Android’s File-Based Encryption

This research [13] investigates the amount of information leakage through unencrypted metadata in Android’s file-based encryption (FBE), which entered as an alternative to the full-disk encryption (FDE) in Android 7.0. FBE allows direct boot, meaning the user can access encrypted data on the lock screen without providing the credentials straight away. This paper shows that despite data encryption, this new method leaves a significant amount of metadata unencrypted. They also show that information leakage through metadata can be used to reconstruct a user’s behavior, which is what we’ve discussed in previous paragraphs (3.2).

Methodology They developed a method based on the IT forensics toolchain *sleuthkit*, which allows them to extract metadata from ext4 partitions; and then from the metadata they collected they propose a method to draft a list of all the installed apps on the device. To reconstruct the metadata they analyzed the static files, which do not change after installation (such as the base.apk file size and native libraries), and used these as the apps fingerprints, they were able to identify 3903 unique applications. To track the user behavior they developed a linking tool which identifies dynamic data, fundamental to reconstruct user actions, and associates it to the application static folder. They were able to achieve this by using the behavior of generation IDs, which are assigned and incremented when folders are created during an app’s installation. To then reconstruct users behavior they perform cycles to check for changes in the metadata: after every cycle their tool returns a list of new, changed, modified and deleted files. Then compare the data which timeframes data and determine exactly when certain actions occurred.

WhatsApp User Action Reconstruction The researchers conducted a case study to demonstrate that they could reconstruct users’ actions in WhatsApp using only unencrypted metadata. Even though they could not identify characteristic fingerprints for specific messaging actions, they discovered that for every action WhatsApp’s databases get updated timestamps. They could identify database files based on the predictable order in which WhatsApp creates them after installation and first use. They clustered their unknown database files by creation time and generation IDs and then knowing the creation patterns, they could identify each database file. This way they were able to trace the exact time a user sent or received a text message or a photo.

The paper concludes that FBE is more insecure than FDE, sacrificing a degree of privacy for user convenience; but they also specified that this vulnerability is limited to devices that run Android 7.0 and 8.0, because with Android 9.0 google introduced metadata encryption which mitigated the flaws exposed in their research.

4.2 FL is Not Private for Google GBoard Next Word Prediction

Federated Learning (FL) is a class of algorithms for the training of machine learning models, these algorithms are trained collaboratively across multiple devices with out exchanging raw data. FL is designed to protect users' privacy, but in this study [14] demonstrate that FL is not private for Next Word Prediction and show how a server can reconstruct the exact words and phrases that a user has typed.

This paper focuses on Gboard's Next Word Prediction (NWP) system, that works on a Long Short-Term Memory (LSTM) recurrent neural network. The threat model consists of an attacker that has access to the FL model architecture and the FL model parameters, the attacker attempts to gain information from the incoming data. In these NWP models the output is naturally similar to the input, so it directly leaks data.

The reconstruction attack consists of two stages: the word recovery stage compares the global parameters with the user local parameters and outputs a gradient per each word, positive if the user did not type the word, negative if the user did type them. This way the negative values reveal the set of words used during the local training session.

The second stage is where the sequences are reconstructed, the attacker feeds to the first word of a sentence to the model and generates a sentence filtering out the words that aren't part of the extracted vocabulary from the previous stage. This process consist of many permutations in order to find the true typed sentences, the attacker measure the percentage drop in perplexity of the phrases, the one with higher perplexity reduction represent the user's typed phrases.

The simulated an attack against Vanilla FL, they achieved near perfect F1 scores stating that it remains highly effective even when local training sizes are scaled up to 256 sentences. The threat becomes more relevant when a real identity can be linked to the sentences and this can happen with telemetry data. Google telemetry often tags updates with an Android ID, then the reconstructed privacy sentences can be linked to a user's real world identity via Google play Services, SIM or linked credit cards.

Eventually they conclude that if the architecture evolves privacy is not entirely lost, because the reconstructed text is redacted, words out of the vocabulary are masked; and also changing from a word based text model to a character based one would make attacks much harder to perform.

4.3 Secure Military Communication

This study [15] aims to develop a prototype system for the secure exchange of messages between users of workstations connected with the same TCP/IP networks. This system is desined to meet high-security operational requirements of a military unit, assuring confidentiality and read only by authorized personnel. This system is based on maintaining high-level security while remaining practical for real world deployments.

The data confidentiality is based on the Advanced Encryption Standard (AES) and they use symmetric encryption, because of its high speed and efficiency. The most important part of symmetric encryption is the key management, it is the hardest and most error-prone part

of secure communication. In this study the authors focus on building a secure lifecycle management system for secret keys, this system automates the process and hides its complexity to the end users.

The management of secret key consist of various steps and security layers: it begins with user registration and initialization, this consists of allowing an entity to become an authorized member of a security domain using a initial key and then initializing the cryptographic application.

The next step consist of generating the key, during this process the algorithm must ensure randomness when creating keys, for them to be unpredictable. Then it needs to be installed and it is done manually or with token passphrases, and along with that it needs to be registered and so associated to a unique name/entity.

The final part of the process is the use and update of the key; the system automates updating and refreshing operational keys before the original keys reach their expiration period, reducing their vulnerability window.

This application basically operates as a specialized peer-to-peer secure chat infrastructure, rather than relying on centralized third-party structures, it allows decentralized encrypted pairing. This application is meant not only as a standalone tool but as the foundation for the development of an integrated, large-scale communication framework which is able to bridge multiple sites across an entire military or high security commercial organizations.

Now even though this study emphasizes the security in the management of AES keys, if the architecture is not encrypted, it can still leak metadata during the exchange; the attacker may not be able to translate the cypher text but it can still identify the sender's identity and the type of message their exchanging.

5 Discussion about mitigations/protectoins/applications

In this section we will analyze some applications of innovative methodologies that aim to provide more security. The main methodologies that have been introduced in this research are based on static encryption, but with traffic analysis and real time reaction to an attack the level of defense can change drastically. In the next paragraphs we will analyze a few new approaches that provide a more adaptive defense system that results to be more effective in nowadays networks.

5.1 Network Attack and Defense Stochastic Differential Game Model

In this study [16] the authors construct a stochastic evolutionary game model based on the stochastic differential equation. In this attack and defense model the groups choose different strategies based on the incomplete rationality of attackers and defenders. Both attackers and defenders constantly adjust and improve their methods. Both attacker and defenders had a set of strategies, meaning potential moves, and payoffs that represented gains and losses in different scenarios. Payoffs are calculated based on distinct parameters: costs before and after the attack, probability of success before and after, defenders original assets and increased defensive investment, social rewards and the actual loss suffered.

The core of this study is modifying traditional replication of dynamic equations by adding a white noise (Wiener) process that represent environmental uncertainty. The study proves that this set of equations outputs a unique solution and obtains the evolutionary equilibrium of attack and defense.

Simulation results and analysis They simulated the model using the explicit Euler numerical method on a practical network topology that consisted of an attacking host, a firewall, a router and three intranet servers (MySQL, Web and FTP).

They analyzed how initial probabilities of attackers (x) and defenders (y) evolved over time, in different conditions. When defenders are actively willing to invest ($y=0.7$) and the attackers are more passive and moderately aggressive ($x=0.4$); the system stabilizes quickly dropping the attacker probability to 0 while the defense probability rises to 1, this shows the creation of a secure environment.

Another possible scenario is when both attacker and defenders start with high-moderate probabilities ($x=0.5$, $y=0.6$), in this case both statuses converge to 1, this represent a network where both attack and defense are active and the network environment is in a relatively fierce state.

A problematic situation arises when defenders start from a low probability ($y=0.2$) and the attackers with a high probability ($x=0.7$), in this case the experimental results show that the attackers probability tends to 1 and the defenders gradually approaches 0.

The study concludes that the return on security investment of their model is better than other related kinds of literature, their model is more realistic than traditional static models.

5.2 AI powered Dynamic Encryption and Decryption defense model

Traditional cybersecurity systems usually use static encryption, this means they rely on fixed algorithms and predetermined keys without any regard to what happens on the network. In this article [17] the authors believe that this approach is ineffective nowadays adaptive, AI-driven cyber threats.

The dynamic Methodology They propose a more dynamic defense model: it is a automated cybersecurity pipeline that connects threat detection directly to a proportional cryptographic response. Encryption dynamically changes based on the severity of the threats that the system is facing. The framework consists of different stages, it begins with the detection stage which starts with Suricata, an open source intrusion detection system, monitoring traffic and producing detailed alert logs (eve.json). These logs are then analyzed by a Random Forest classifier that assign to each 1 of them 1 of the four security levels (Low, Medium, High, Critical).

Next it is used the dynamic encryption: based on the severity of the threat, a weighted random function dynamically selects and encryption scheme, for higher threat the function chooses stronger or faster algorithms to protects the files but stills manages the computer processing efficiency.

To enable continuous operation, the system uses Windows Task Scheduler that automates this workflow to run every three hours and it uses failure-recovery logic to prevent system crashes or overlapping tasks. Decryption is manually triggered by authorized users and it is only limited to a five-minute read only window, after which the file is automatically re-encrypted. All operations, such results, encryption type, decryption attempts and access duration are logged to support forensics investigations.

Findings and Results The framework was tested and evaluated in a controlled environment; they found that the Random Forest algorithm had a 79% level of accuracy in determining the security level. While it easily identified low risk traffic, it struggled to differentiate between classes of adjacent levels, mostly due to class imbalance. The system maintained a 93.3% success rate over 30 runs, with an average encryption latency of 5-9 ms per log.

The study concludes that integrating a ML-based threat classification with adaptive encryption assignment demonstrated strong modularity, automation and forensics traceability. It provides an excellent framework for enterprise and cloud environments and removes human delay while maintaining audit trails.

6 Conclusion

This research evaluates network encryption and traffic analysis from many points of view; we have demonstrated that even though modern encryption protocols, like TLS and IPsec, are fundamental to data confidentiality, they are not sufficient to guarantee complete protection. The main vulnerability is what encryption leaves behind and that would be the metadata: it can provide a rich vector of information such as packet sizes, directional flows and timing that an attacker can use to reconstruct user behavior, classify encrypted traffic from applications and decode sensitive communication protocols.

With the examples examined, we can map the widespread of the vulnerabilities from everyday life with the Android’s FBE and Google’s GBoard NWP to national defense frameworks with the military encryption discussion. The leak of sensitive information from the metadata affects every data that is on the network, and especially in military networks it is a vulnerability that needs to be assessed. Encryption alone is not enough to protect military networks and the leak of operational data can generate dangerous consequences.

We have come to the conclusion that in order to combat these highly adaptive, AI driven attacks, cyber defense must shift from static models to more dynamic ones. As discussed, transforming defense into an automated, proactive mechanism that dynamically adapts to the real-time threat can mitigate the vulnerability. Lastly its necessary to ensure an integrated, multi-layer approach that not only secures the payload but also masks, obfuscates and dynamically defends the metadata.

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